PART 2 - CAPSTONE PROJECT #6

Python Simulation Code & Waveforms

Task 65: Verified Google Colab Python Script for AM, FM, and ASK.

1. Python Code (AM & FM Simulation)

```
      import numpy as np import matplotlib.pyplot as plt t = np.linspace(0, 1, 1000) fm, fc = 5, 50 Am, Ac = 1.0, 2.0 m_t = Am * np.cos(2 * np.pi * fm * t) s_am = (Ac + 0.5 * m_t) * np.cos(2 * np.pi * fc * t) s_fm = Ac * np.cos(2 * np.pi * fc * t + 5.0 * np.sin(2 * np.pi * fm * t))
```

2. Simulated Waveforms



3. Digital ASK Python Code

```
bits = np.array([1, 0, 1, 1, 0, 1]) digital_signal = np.repeat(bits, 100) t_digital = np.linspace(0, len(bits), len(digital_signal)) carrier = np.sin(2 * np.pi * 5 * t_digital) s_ask = digital_signal * carrier
```

4. Digital ASK Waveform

